

碳达峰的一般规律、必要条件与影响因素的边际贡献



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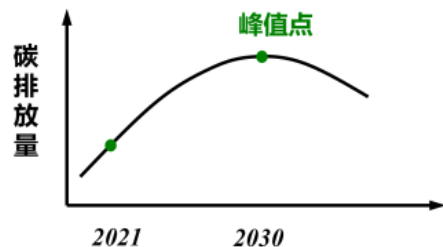


一、什么是“双碳”？



Peak Carbon Dioxide Emissions

在某一个时点，二氧化碳排放不再增长达到峰值，之后逐步回落。



Carbon Neutrality

与某一主体相关的**人为**二氧化碳排放量与**人为**二氧化碳清除量相平衡的状态。

IPCC 定义

净零排放 (Net zero CO₂ emissions):

在全球层面，与**碳中和**等价



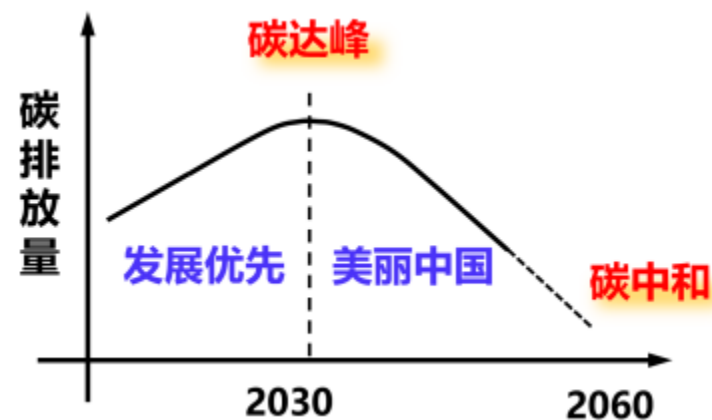
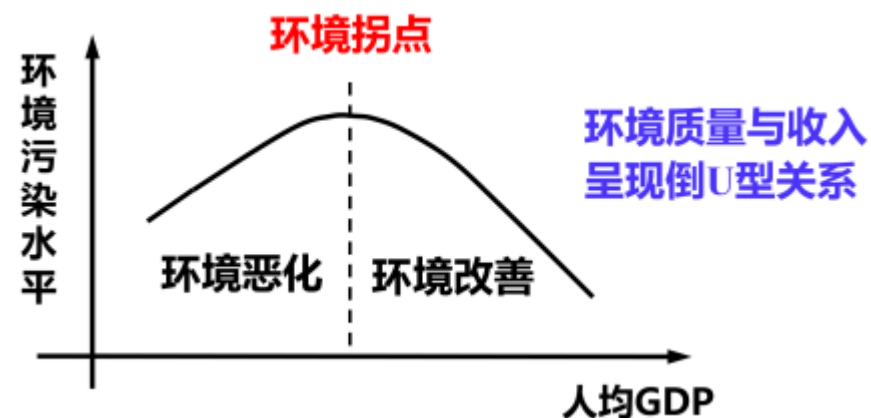
排放 = 吸收

双碳目标的经济理论基础

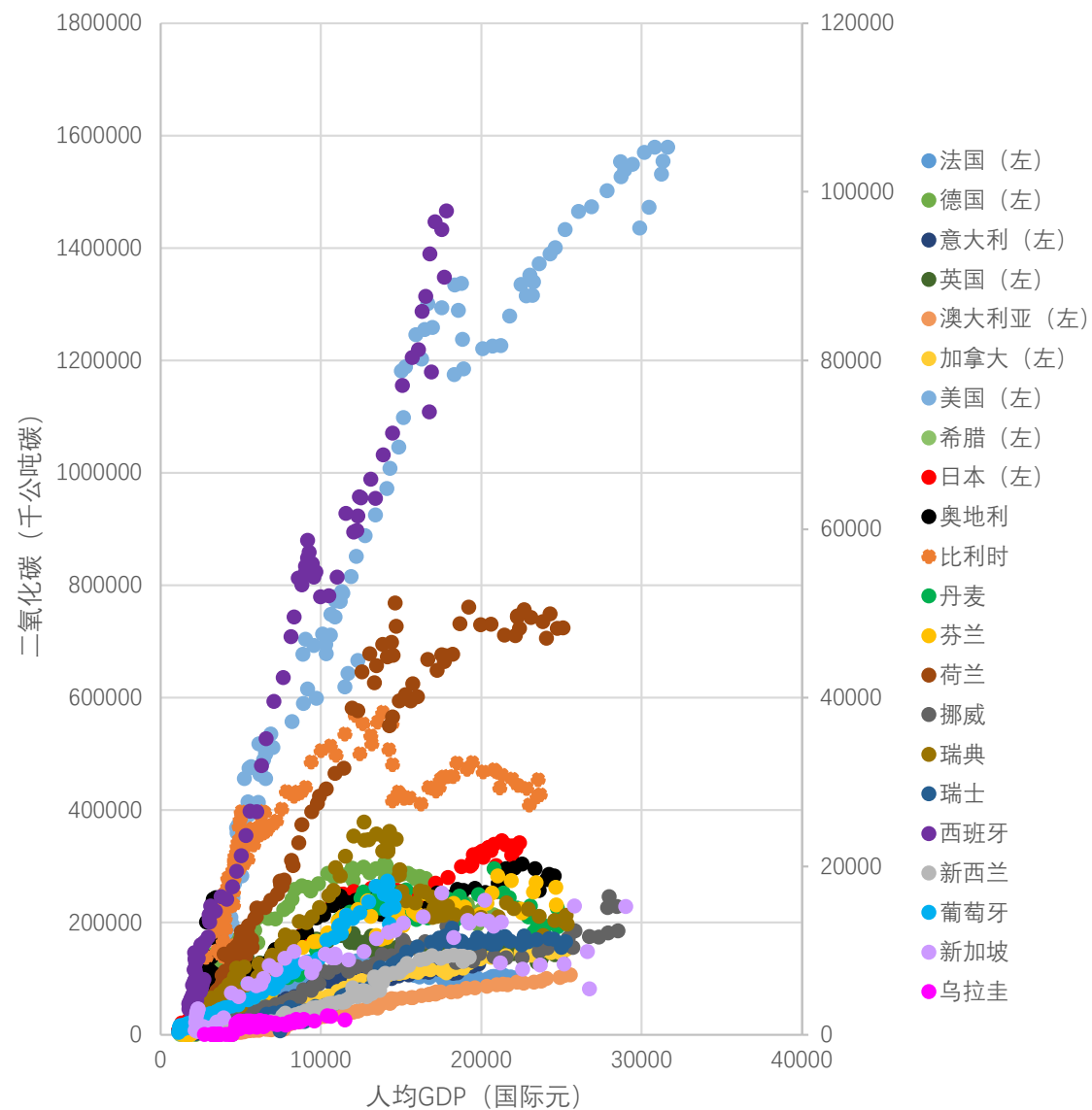
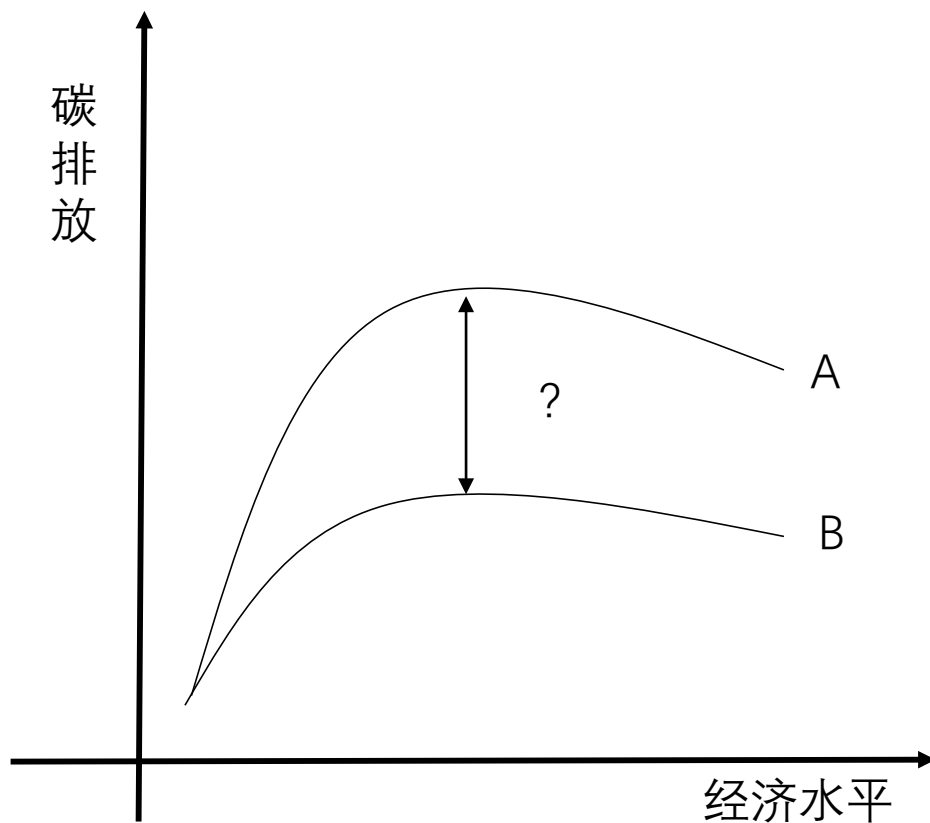
根据环境库兹涅茨曲线理论，环境质量与经济发展水平呈现倒U型关系；随着经济发展，一国的二氧化碳排放最终将达到峰值并开始下降。但若不加以政策干预，这一过程可能持续数百年。

在当前全球气候变化的严峻挑战下，我国要实现经济发展与碳减排目标的协同发展，相当于“戴着镣铐跳舞”。

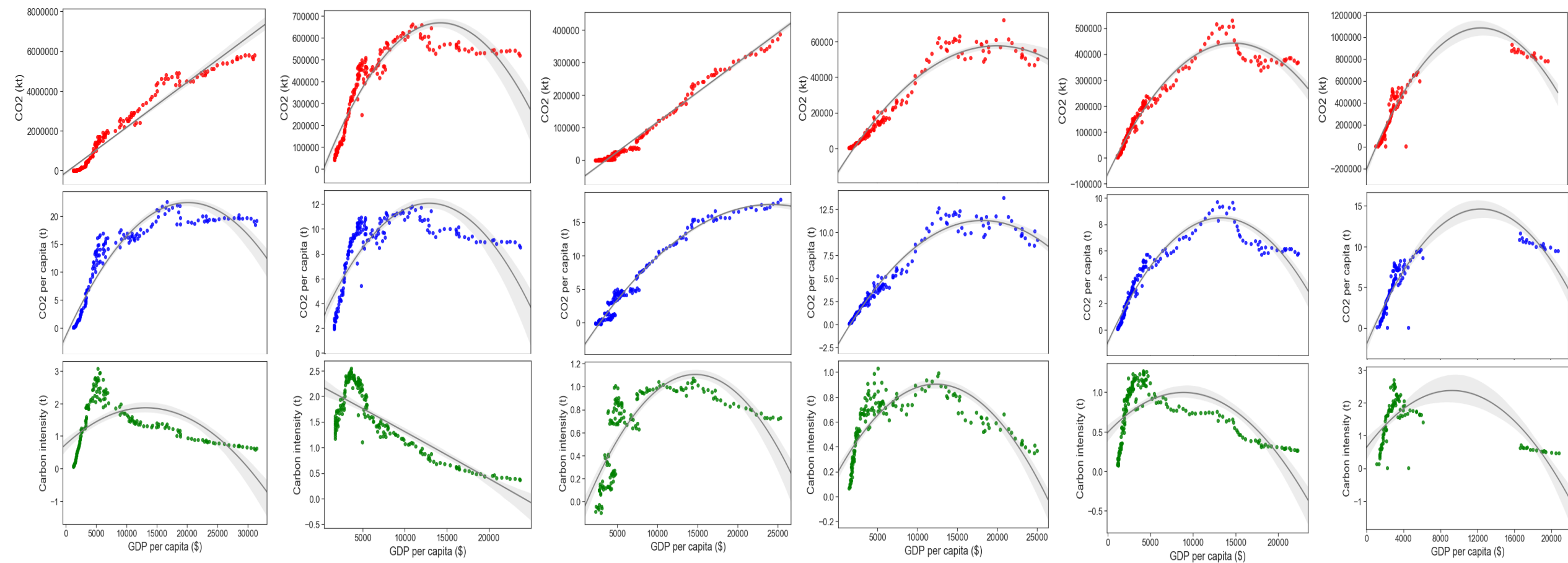
如何更早、更好地完成碳减排目标成为亟待解决的问题。



二、问题的提出



三、达峰一般规律



USA

UK

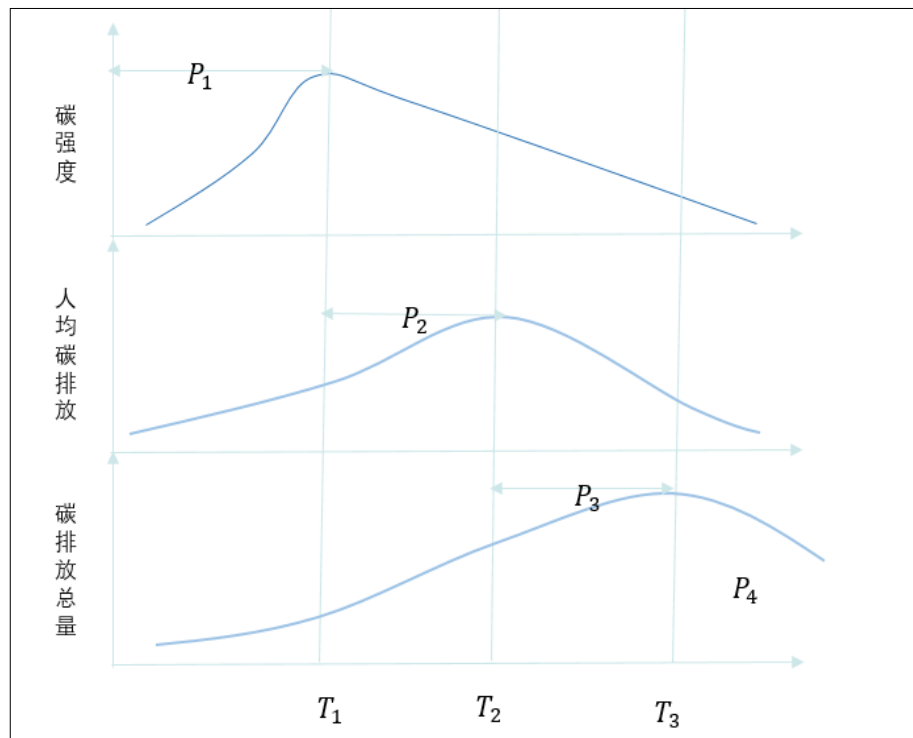
Australia

DENMARK

FRANCE

GERMANY

三、达峰一般规律与必要条件



命题1：人均碳达峰的必要条件是碳强度的指数下降率要大于人口指数变化率与人均GDP指数增长率之和。即

$$-(\ln \frac{CO_2}{GDP})' \geq (\ln \frac{GDP}{P})'$$

命题2：碳达峰的必要条件是碳强度的指数下降率要大于人口指数变化率与人均GDP指数增长率之和。即

$$-(\ln \frac{CO_2}{GDP})' \geq (\ln P)' + (\ln \frac{GDP}{P})'$$

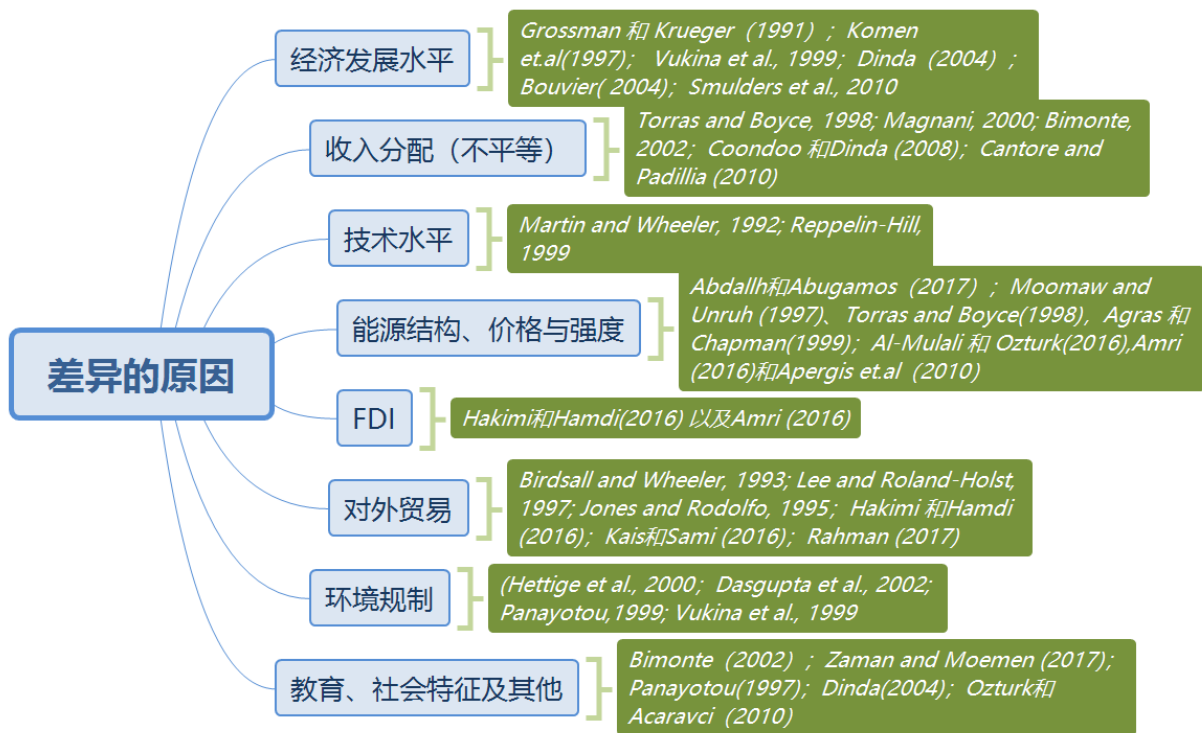
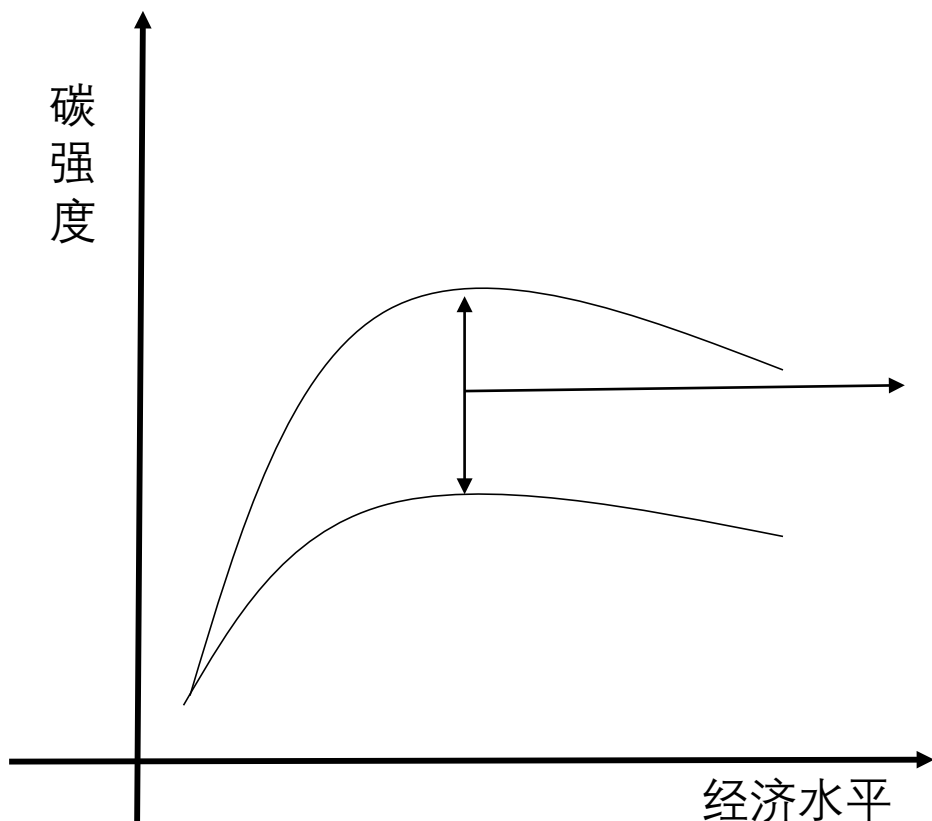
三、达峰一般规律与必要条件

$$CO_2 = P \cdot \frac{GDP}{P} \cdot \frac{CO_2}{GDP}$$

对式两边同时对时间同时求导数，可有：

$$\begin{aligned} CO_2' &= P' \cdot \frac{GDP}{P} \cdot \frac{CO_2}{GDP} + P \cdot \left(\frac{GDP}{P}\right)' \cdot \frac{CO_2}{GDP} + P \cdot \frac{GDP}{P} \cdot \left(\frac{CO_2}{GDP}\right)' \\ &\Leftrightarrow CO_2' = P' \cdot \frac{CO_2}{P} + \left(\frac{GDP}{P}\right)' \cdot \frac{P \cdot CO_2}{GDP} + \left(\frac{CO_2}{GDP}\right)' \cdot GDP \\ &\Leftrightarrow \frac{CO_2'}{CO_2} = \frac{P'}{P} + \frac{\left(\frac{GDP}{P}\right)'}{\frac{GDP}{P}} + \frac{\left(\frac{CO_2}{GDP}\right)'}{\frac{CO_2}{GDP}} \\ &\Leftrightarrow (\ln CO_2)' = (\ln P)' + \left(\ln \frac{GDP}{P}\right)' + \left(\ln \frac{CO_2}{GDP}\right)' \end{aligned}$$

四、碳排放的影响因素

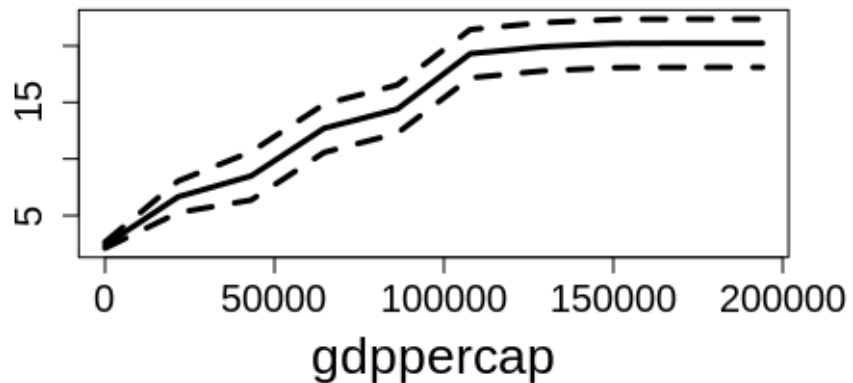


五、变量重要性（从高到低排序）

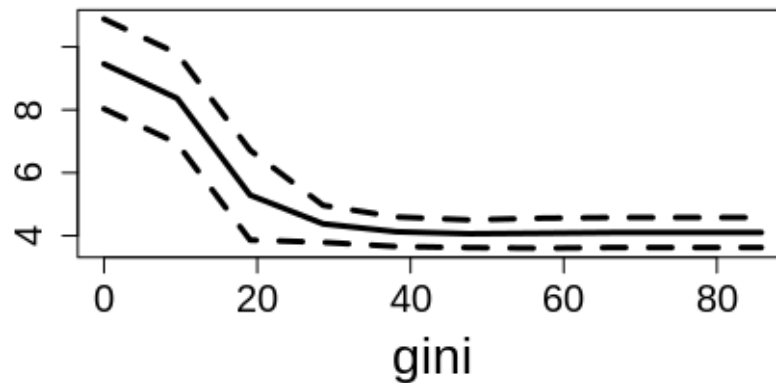
	Predictor	Marginalized error	Model error	Relative change	Z-value
7	gdppercap	40.5727	3.6149	10.224	13.041
10	gini	21.1822	3.6149	4.860	10.230
15	gasoline	19.7109	3.6149	4.453	10.867
8	rd	19.1418	3.6149	4.295	10.675
9	edu	18.9734	3.6149	4.249	11.325
14	deisel	18.4310	3.6149	4.099	11.062
5	export	18.3113	3.6149	4.065	11.972
6	import	17.9544	3.6149	3.967	11.372
17	infopub	17.9388	3.6149	3.962	11.133
19	oilconsumption	17.9376	3.6149	3.962	11.206
16	envpolicy	17.9306	3.6149	3.960	11.244
18	enerintens	17.9200	3.6149	3.957	11.197
4	service	17.8953	3.6149	3.950	11.185
13	info	17.8815	3.6149	3.947	11.169
11	fdi	17.8520	3.6149	3.938	11.187
1	Country	17.8146	3.6149	3.928	11.186
2	time	17.8146	3.6149	3.928	11.186
20	CO2percap	17.8146	3.6149	3.928	11.186
3	manufacture	17.8106	3.6149	3.927	11.206
12	property	17.7855	3.6149	3.920	11.253

六、边际效应

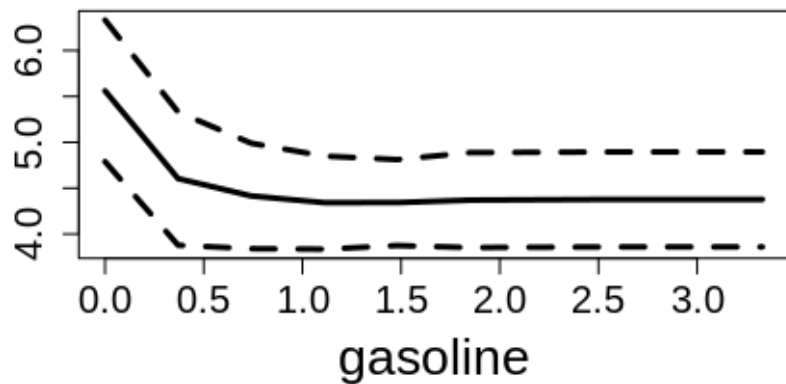
Marginal effect



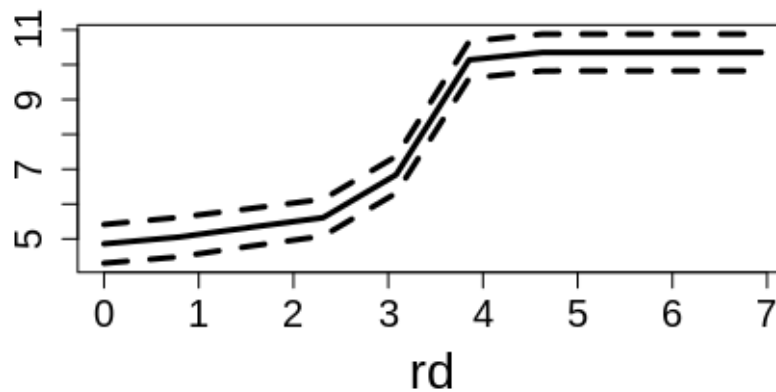
Marginal effect



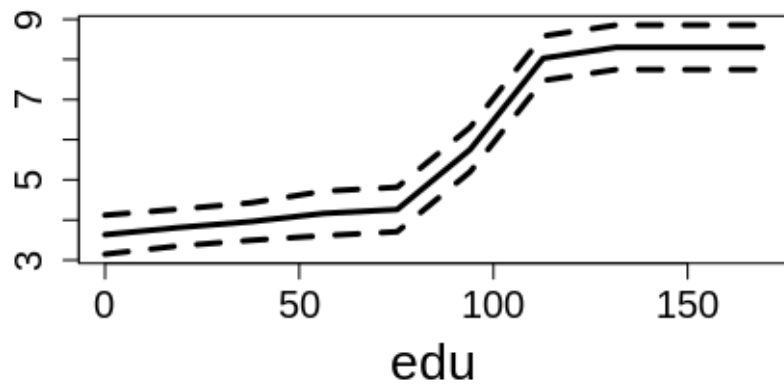
Marginal effect



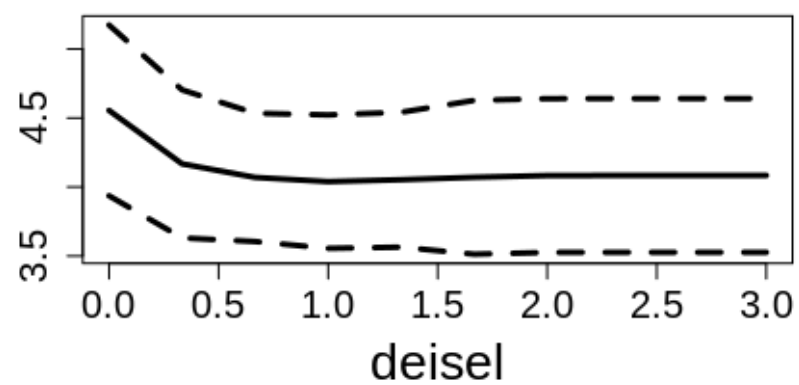
Marginal effect



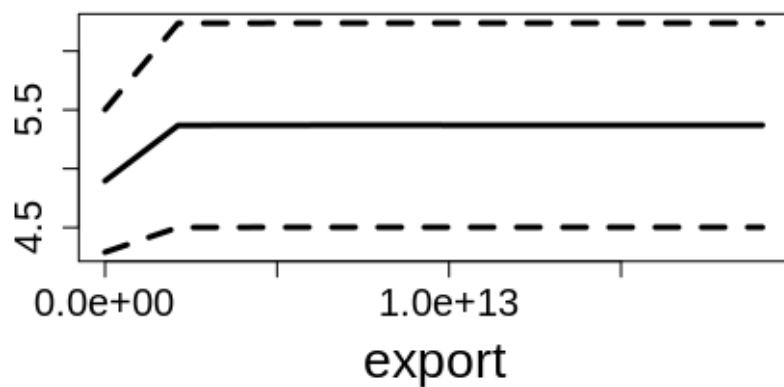
Marginal effect



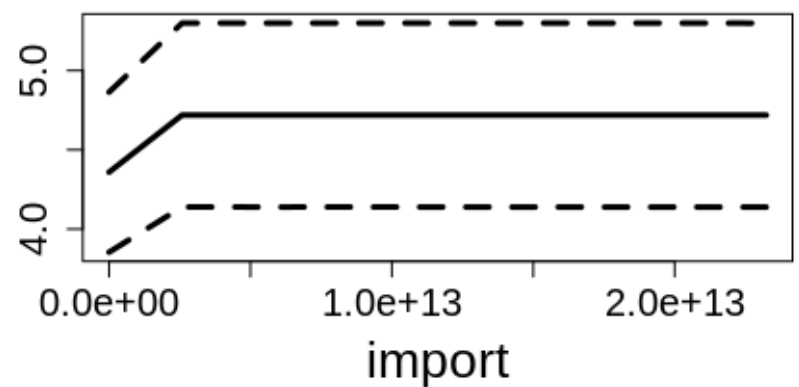
Marginal effect



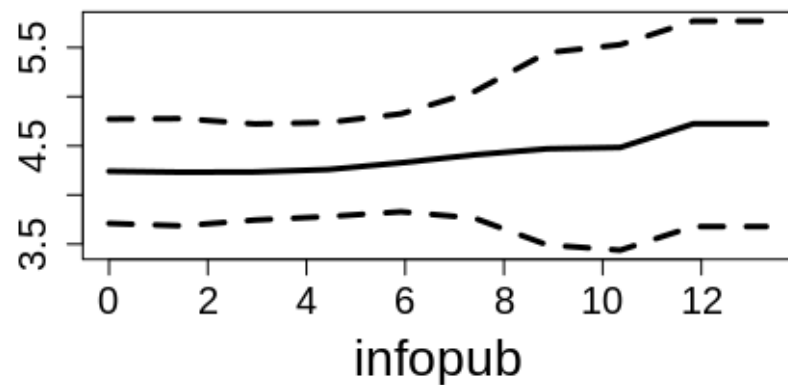
Marginal effect



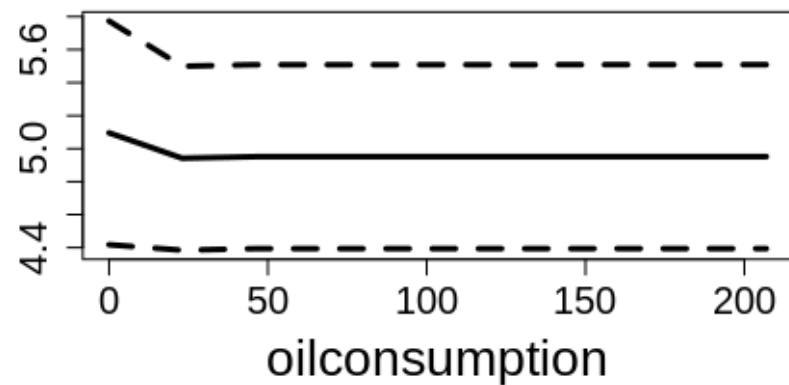
Marginal effect



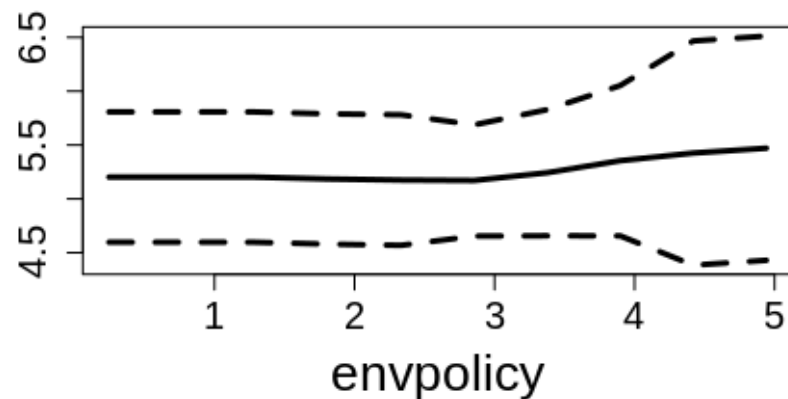
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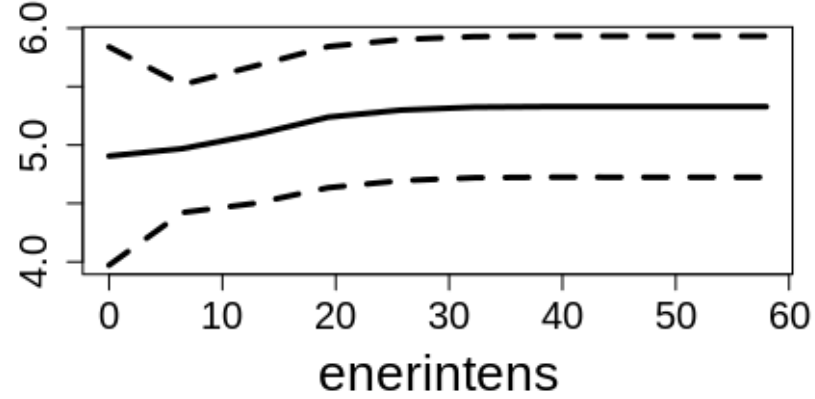
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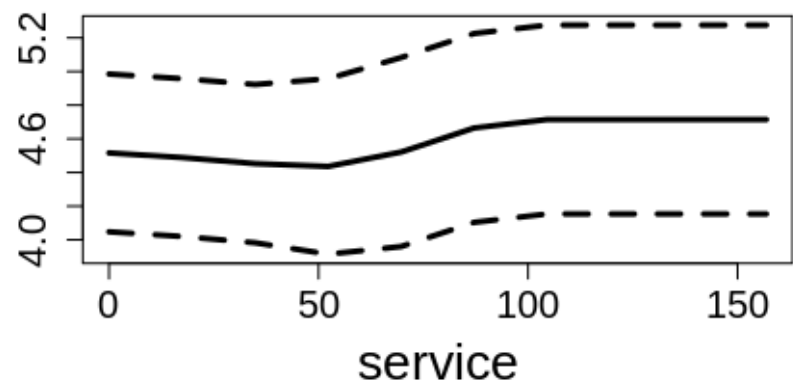
Marginal effect



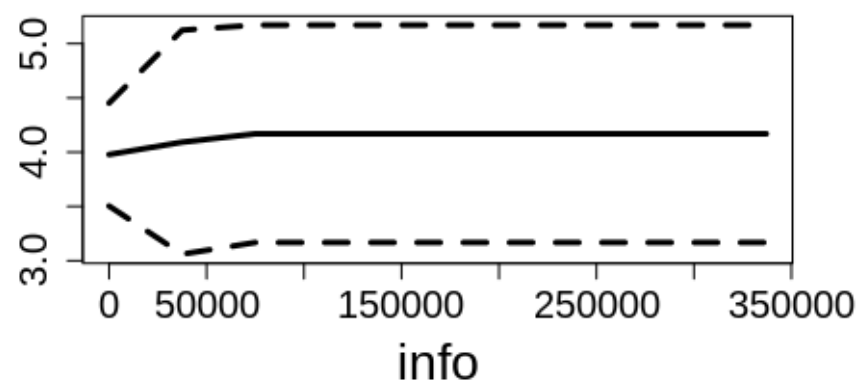
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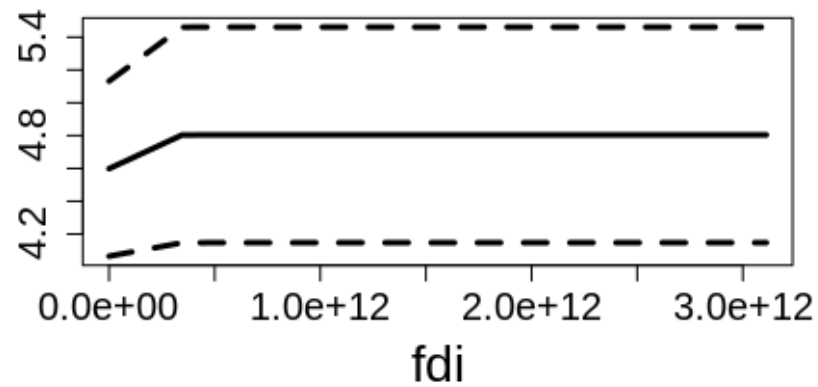
Marginal effect



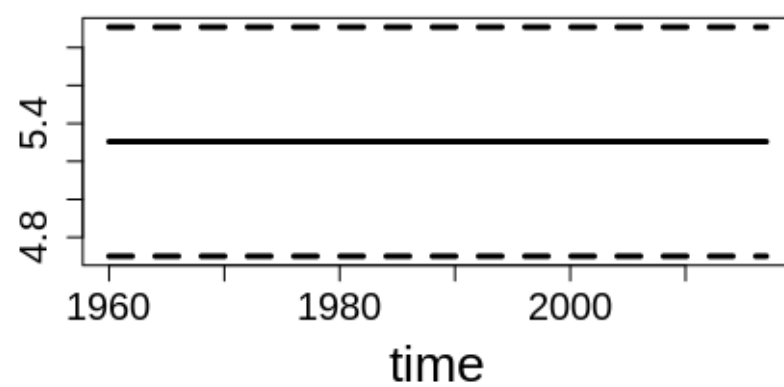
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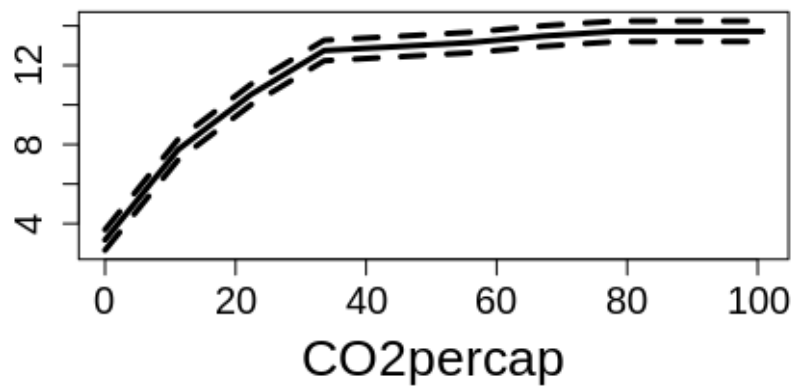
Marginal effect



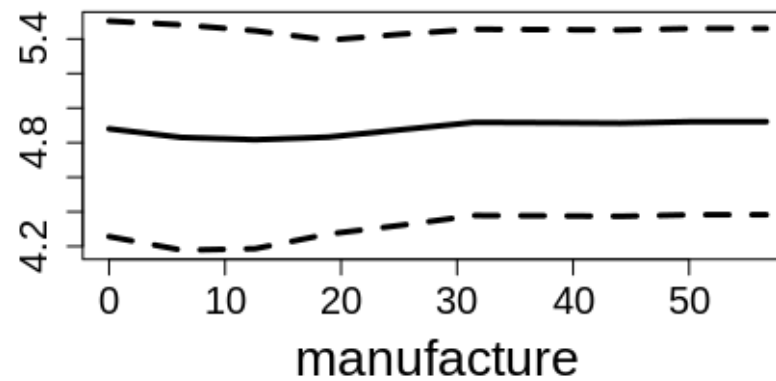
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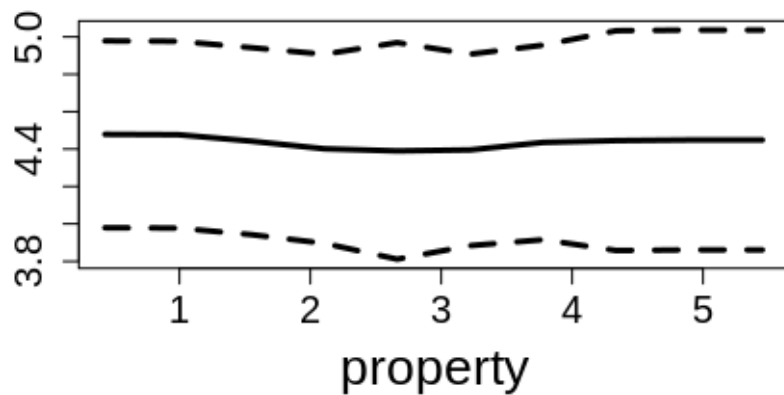
Marginal effect



Marginal effect

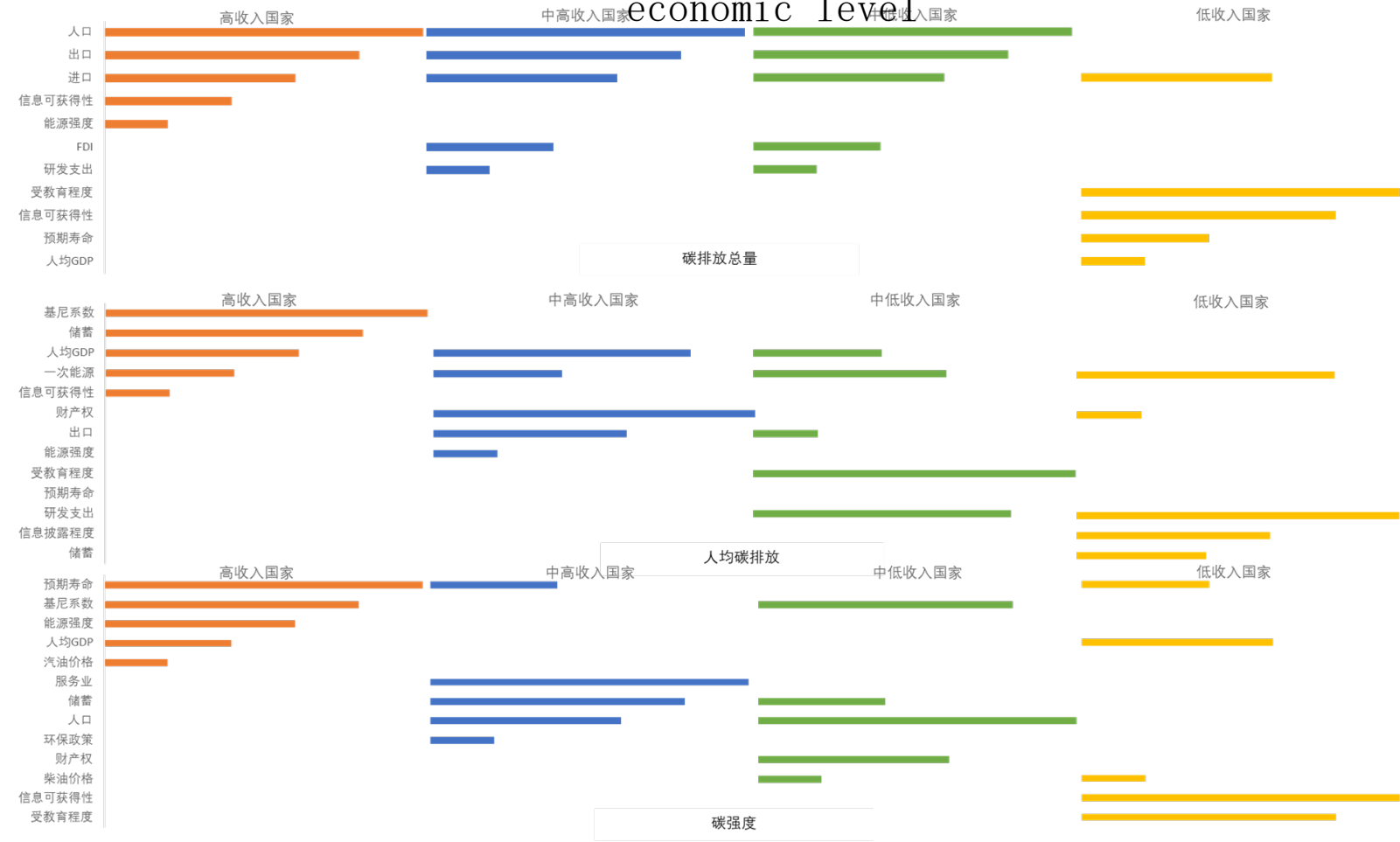


Marginal effect

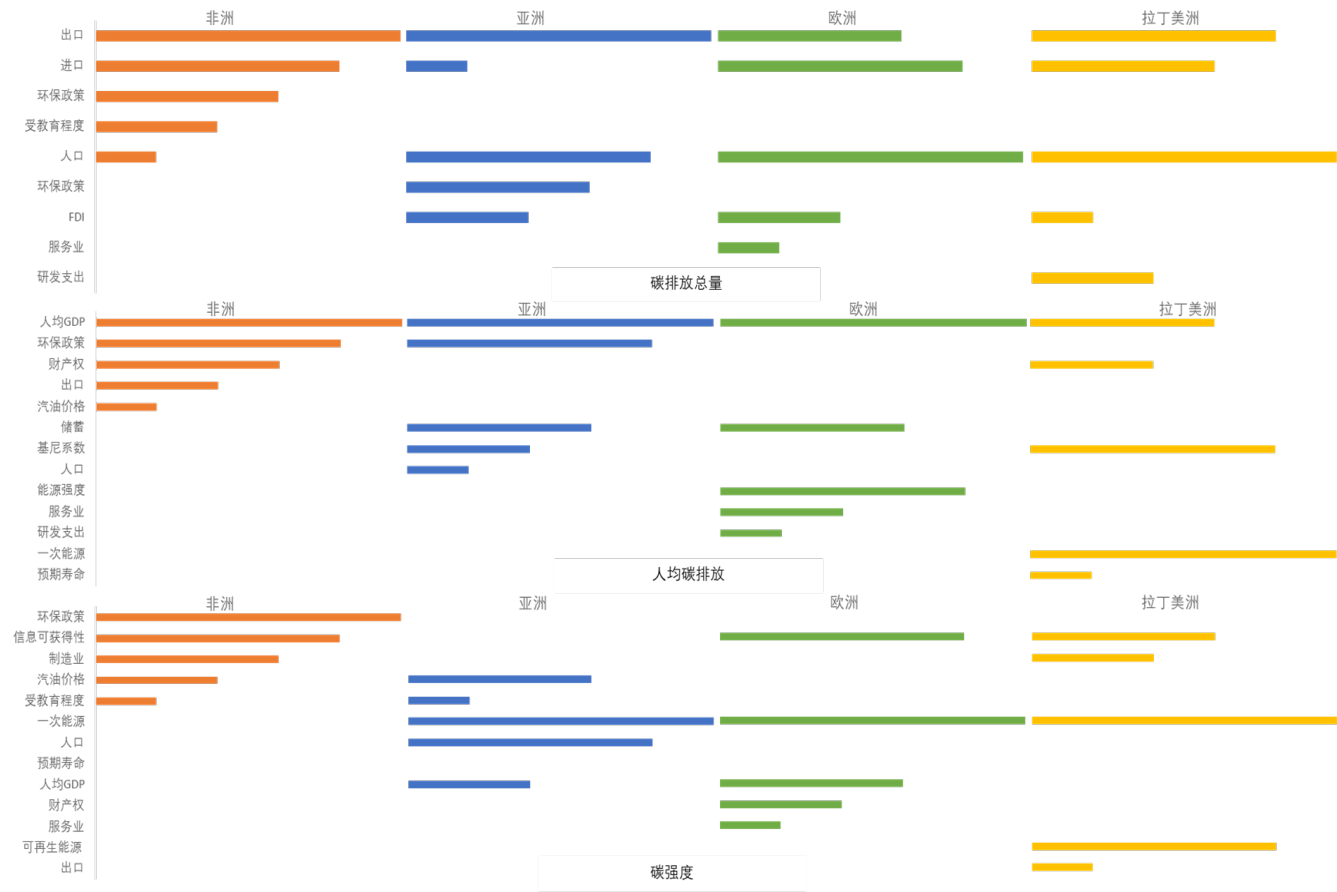


七、进一步分析

Top 5 factors of CO₂ emission in areas of different economic level



Top5 factors of CO₂ emission in different geographic areas



八、结论与讨论

(1) 控制碳达峰的重要政策工具是碳强度，也是协调经济发展与碳减排的主要工具；

(2) 影响碳排放的重要影响因素具有时空异质性，各地应根据实际情况对政策进行动态调整；

(3) 注意政策组合的作用。